Compliance sheet — star-ecliptic

Everything a supplier qualification, a legal review or an export check usually asks about star-ecliptic 0.1.0, in one document.

1. Identity

Document	STAR-PKG-STAR-ECLIPTIC-EN · v1.2 · 2026-10-15
Package	star-ecliptic
Version	0.1.0
Capability	star.frames.ecliptic
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_ecliptic

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE and NOTICE files, keep the copyright notices, and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_ecliptic-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_ecliptic-0.1.0-py3-none-any.whl	a73d57d15622e15c3ff2ab2ea8eb5eb355e54263f5d677891e2cd9188a33b21f
star_ecliptic.py	6ce870a4af69d5fba1bd955150b5c1418b82bc66a818f61237291fd749d70d6e
star_ecliptic-0.1.0-source.zip	45c02ca7f9cd68f0d2c6c208916d92f7b76d4efadda90008652c642acde595ac

5. What it does, and its limits

Mean obliquity of the ecliptic (IAU 2006 and IAU 1980) and conversion of directions between equatorial and ecliptic frames; agrees with ERFA, astropy and SPICE to 1.1e-13 deg and with ERFA and skyfield to 1.5e-11 arcsec on 600 cases, and refuses out-of-range input.

Mean obliquity only: no nutation (pass a true obliquity if you have one). No precession: the directions stay referred to the equinox they were given in. The IAU 1980 polynomial is compared with one library (ERFA) and one published example, not two libraries. At a pole of the target frame the longitude is undefined and the value returned there is arbitrary (within [0, 360)). The time argument is TT; using UTC instead changes the result by less than 1e-6 arcsec.

- `mean_obliquity_arcsec(jd_day, jd_frac, model)` evaluates the IAU 2006 (default) or IAU 1980 polynomial in TT Julian centuries from J2000.0, with the date split in day and fraction. Reproduces Meeus Example 22.a.
- `equatorial_to_ecliptic` and `ecliptic_to_equatorial` are one rotation about the equinox by the obliquity passed by the caller, and are inverse of each other; longitudes out are in [0, 360). Reproduce Meeus Example 13.a.
- Measured on 600 cases (dates 1800–2200, 100 directions down to 1e-6 deg from the pole of the ecliptic): directions within 1.1e-13 deg of ERFA, astropy and SPICE; IAU 2006 obliquity within 1.5e-11 arcsec of ERFA and skyfield; IAU 1980 obliquity within 1.5e-11 arcsec of ERFA.
- Every function returns finite floats or raises `ValueError`: non-numeric, boolean, NaN or infinite input, a date outside JD 2086302.5 … 2816787.5 (years 1000–3000), a fraction outside [-1, 1], an

unknown model, declination or latitude outside [-90, 90], right ascension or longitude outside [-360, 360], obliquity outside [0, 324000] arcsec.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_ecliptic_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-ecliptic}: Mean obliquity and equatorial ⇄ ecliptic directions},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_ecliptic/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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